Quantification: 548_hyp_HP_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	16
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3029.88	2908.5	2402.63	px^2
major_axis_length_px	lacuna shape	88.2472	92.5844	40.8446	px
minor_axis_length_px	lacuna shape	43.4803	46.7801	21.3602	px
aspect_ratio	lacuna shape	2.2046	1.8291	1.0075	unitless
eccentricity	lacuna shape	0.8261	0.8372	0.1147	unitless
solidity	lacuna shape	0.8018	0.8106	0.0833	unitless
roots_count	headline	10.6875	8	6.809	count
ring_length_r30_px	headline	333.375	313	170.943	px
ring_length_r60_px	reported	614.125	580.5	278.261	px
owned_length_px	ownership-dependent	846.085	797.696	632.324	px
edge_count	ownership-dependent	41.75	42	30.5865	count
mean_edge_length_px	ownership-dependent	19.9826	19.3972	3.7649	px
perimeter_px	normalised	282.406	271.392	148.427	px
ring_area_r30_px2	normalised	9270.31	10118	3842.47	px^2
ring_area_r60_px2	normalised	19979.1	22370	7614.66	px^2
in_frame_fraction_r30	normalised	0.9832	1	0.0412	unitless
in_frame_fraction_r60	normalised	0.9548	1	0.0856	unitless
ring_density_r30	normalised	0.0357425	0.0375812	0.00968205	px^-1
ring_density_r60	normalised	0.0312748	0.0314364	0.00818299	px^-1
roots_per_100px_perimeter	normalised	3.7508	3.9664	1.0597	per 100 px
ring_attached_length_r30_px	network v2	297.125	278	170.818	px
ring_attached_length_r60_px	network v2	448.75	441	262.379	px
ring_length_w_r30_px	network v2	381.502	355.649	194.713	px
ring_length_w_r60_px	network v2	703.241	665.248	316.159	px
sholl_crossings_r10	network v2	8.1875	7	4.2617	count
sholl_crossings_r20	network v2	8.6875	8	4.4977	count
sholl_crossings_r30	network v2	8.25	6.5	4.7117	count
loop_count	loops, report only	2	1.5	2.1602	count
loop_edge_share	loops, report only	0.1134	0.0829	0.1194	share
ring_width_mean_r30_px	width	7.4918	7.6239	0.7266	px

Per field
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measure	value	units
analysed_area_px2	984409	px^2
total_skeleton_length_px	27344	px
canalicular_length_density_per_px	0.0277771	px^-1
junction_count	553	count
junction_density_per_px2	0.00056176	px^-2
lacuna_count	21	count

lacunae_per_px2	2.133e-05	px^-2
skeleton_component_count	371	count
mean_component_length_px	73.7035	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0317804	px^-1
field_density_without_flagged_per_px	0.0277771	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	277	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0277771	px^-1
field_length_density_with_unattached_per_px	0.0280585	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	45	count
loop_edge_share	0.1081	share
edge_count	1462	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6875	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27344	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1925	65.1309	45.5827	1.4289	0.7143	0.7606	0.0397	28.706	675.968	True
2	3961	122.826	46.3249	2.6514	0.9261	0.8431	0.4753	53.5506	452.253	False
3	547	38.4822	18.6369	2.0648	0.8749	0.9071	1.3303	28.7715	719.596	False
4	4287	134.045	52.6927	2.5439	0.9195	0.6794	-0.8737	66.8097	864.372	False
5	3031	74.7141	59.7338	1.2508	0.6007	0.7786	-0.0729	198.233	105.492	False
6	1249	97.3283	20.9931	4.6362	0.9765	0.6665	0.811	244.751	234.965	False
7	833	42.3386	25.9256	1.6331	0.7906	0.8976	0.935	290.168	253.631	False
8	8164	134.063	85.066	1.576	0.7729	0.8013	-0.8691	359.846	342.936	False
9	5658	108.902	72.88	1.4943	0.7431	0.7761	-0.0527	409.015	685.903	False
10	1968	65.0915	47.7903	1.362	0.6789	0.7977	-1.2589	372.308	23.5991	True
11	2190	104.344	39.1379	2.6661	0.927	0.7344	0.1875	474.681	15.2301	True
12	2786	87.8405	47.2353	1.8596	0.8431	0.8199	0.6054	469.408	431.352	False
13	4506	98.6377	61.8033	1.596	0.7794	0.8877	-0.0114	577.89	721.96	False
14	749	67.5097	15.6293	4.3194	0.9728	0.8397	0.4961	722.103	490.606	False
15	534	30.0222	23.0801	1.3008	0.6395	0.9287	0.8021	729.066	431.221	False
16	1086	59.0095	32.8062	1.7987	0.8312	0.7126	1.4627	783.261	210.357	False
17	4263	130.21	50.8324	2.5616	0.9207	0.6991	-1.4214	797.485	870.568	False
18	1744	52.1215	44.4583	1.1724	0.522	0.9483	-1.1943	879.793	1002.11	True
19	6366	155.327	60.2608	2.5776	0.9217	0.7519	-1.2656	913.286	921.258	False
20	7862	142.695	80.4282	1.7742	0.826	0.8281	1.2173	982.477	693.681	True
21	458	30.6993	21.7842	1.4092	0.7046	0.8388	0.4457	956.432	222.284	False

Network per lacuna

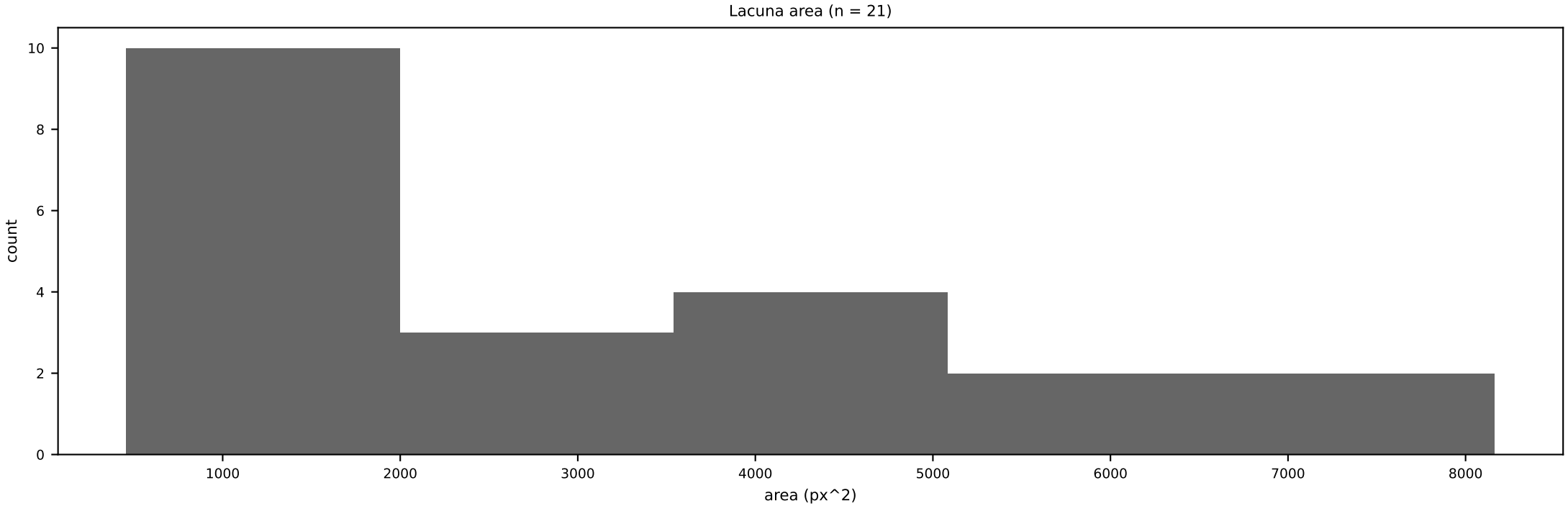
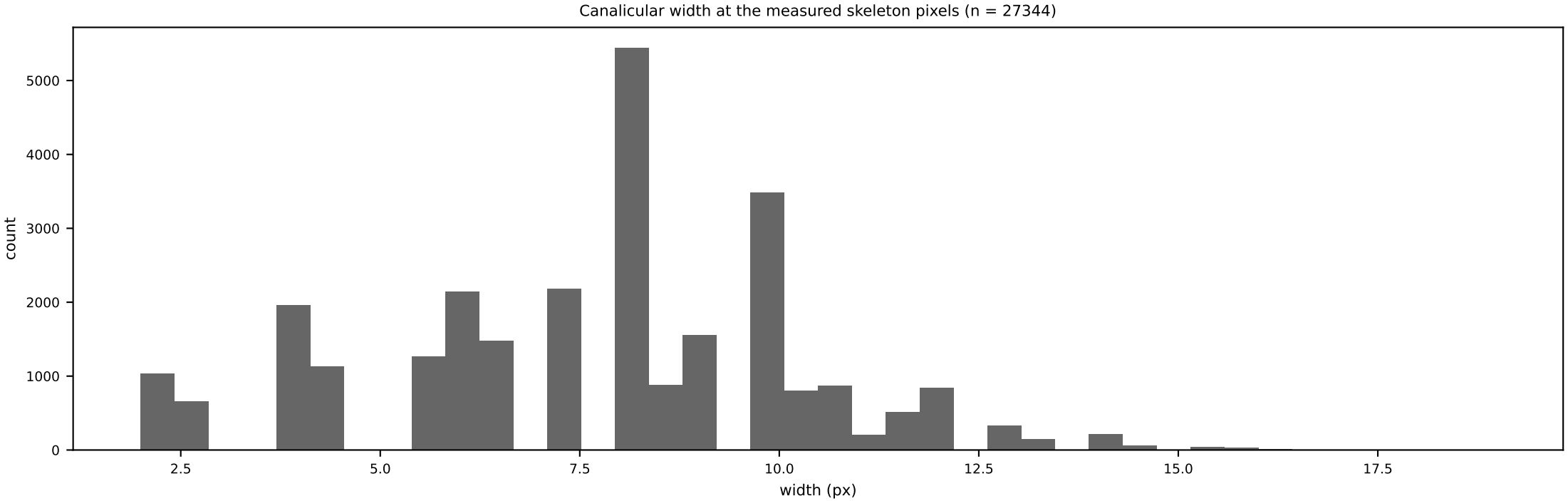
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	196	361	687.34	23	29.8844	6.7244
2	9	251	503	1112.97	68	16.3673	8.3083
3	6	177	283	243.823	16	15.239	6.0179
4	20	530	896	1829.13	87	21.0244	7.6483
5	6	403	827	1056.68	47	22.4825	7.1285
6	10	278	427	273.794	17	16.1055	7.3603
7	5	137	226	197.167	10	19.7167	8.2717
8	25	686	1221	2131.41	108	19.7353	7.7058
9	20	427	776	952.051	50	19.041	7.6251
10	5	227	373	479.505	28	17.1252	7.3694
11	9	167	357	1448.01	70	20.6859	7.5464
12	6	373	723	1361.28	53	25.6846	8.1493
13	13	348	654	1562.55	72	21.7021	7.8628
14	5	194	405	445.031	15	29.6688	7.6226
15	2	80	223	17.0711	1	17.0711	5.8574
16	7	170	496	205.752	11	18.7048	7.3759
17	16	515	885	715.169	43	16.6318	6.9922
18	0	8	50	0	0	0	9.2699
19	16	522	774	880.222	41	21.4688	8.3277
20	9	338	628	562.215	36	15.6171	7.3058
21	5	243	507	553.257	29	19.0778	7.6157

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	220.61	5180	11673	0.7389	0.6782	0.0378378	0.0309261	2.7197
2	327.706	10056	22309	0.8579	0.7712	0.0249602	0.0225469	2.7464
3	102.326	3157	5234	0.9041	0.7441	0.0560659	0.0540696	5.8636
4	434.274	13343	26739	0.9794	0.8718	0.0397212	0.0335091	4.6054
5	274.007	10299	23608	1	1	0.03913	0.0350305	2.1897
6	225.38	6965	13598	1	1	0.0399139	0.0314017	4.437
7	121.983	3984	7252	1	1	0.0343876	0.0311638	4.0989
8	522.985	15465	29728	1	1	0.0443582	0.0410724	4.7803
9	451.63	13578	29430	1	1	0.0314479	0.0263676	4.4284
10	224.368	5419	11920	0.697	0.6419	0.0418897	0.0312919	2.2285
11	356.534	5694	11685	0.6133	0.5772	0.0293291	0.030552	2.5243
12	268.777	10180	22542	1	1	0.0366405	0.0320735	2.2323
13	328.534	11289	25601	1	1	0.0308265	0.0255459	3.957
14	155.681	6604	15865	1	1	0.0293761	0.0255279	3.2117
15	95.6396	4787	11479	1	1	0.0167119	0.0194268	2.0912
16	176.066	7548	20526	1	1	0.0225225	0.0241645	3.9758
17	441.789	13369	27214	1	1	0.0385219	0.03252	3.6216
18	173.397	1958	3433	0.6626	0.6178	0.0040858	0.0145645	0
19	478.902	12272	22431	0.9903	0.9107	0.0425359	0.0345058	3.341
20	502.902	11117	23341	0.7175	0.6712	0.0304039	0.0269055	1.7896
21	112.811	5429	16110	1	0.9788	0.0447596	0.0314711	4.4322

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	173	245	221.995	414.919	5	5	4
2	206	281	282.794	578.63	6	6	4
3	164	206	198.853	324.735	3	4	2

4	459	741	600.387	1034.8	12	10	15
5	336	704	457.747	935.151	8	14	14
6	220	286	315.25	478.375	6	11	5
7	105	164	163.095	261.522	2	5	4
8	663	997	787.311	1390.21	16	20	12
9	385	560	488.345	891.607	11	9	11
10	212	334	252.51	423.605	6	4	6
11	166	307	177.912	395.907	4	5	3
12	348	560	432.718	825.767	14	8	10
13	339	562	396.049	741.713	10	8	13
14	153	274	230.451	480.872	4	8	5
15	54	54	90.1543	253.693	5	2	2
16	127	187	194.338	573.227	5	4	7
17	487	594	587.316	1007.48	12	12	6
18	0	0	8.2426	58.4264	0	0	1
19	502	688	596.144	885.293	12	12	16
20	309	422	388.019	704.813	8	5	7
21	206	322	283.078	588.784	5	6	6

548_hyp_HP_z08c1-2: distributions, pre-validation, px